CHAPTER 3: EQUATION

3.1 Quadratic equation

A quadratic expression is an expression in the form $ax^2 + bx + c.$, where *a, b* and *c* are constants, $a \neq 0$ and *x* is an unknown.

- A quadratic expression satisfies the following conditions:
    a) There is only one unknown
    b) The highest power of the unknown is 2.

- A quadratic equation satisfies the following conditions:
    a) There is only one unknown
    b) The highest power of the unknown is 2
    c) There is an equal sign '='

3.1.1 Solution by factorisation

- Solve an equation' means 'to find the value of the unknown'.
- The value of the unknown is called the root of the equation.

If a quadratic equation can be factorized into a product of two factors such that

$$(x-p)(x-q) = 0,$$

Hence $x - p = 0$ or $x - q = 0$

$x = p$ or $x = q$

$\therefore$ p and q are the roots of the equation .

Notes

1. If $p \neq q$ $\rightarrow$ the equation have two different roots
2. If $p = q$ $\rightarrow$ the equation have two equal roots (one root only)
3. The equation must be written in general form $ax^2 + bx + c = 0$ before factorization.

Solve the following quadratic equations by factorization .

1. $x^2 - 7x - 8 = 0$ $(x-8)(x+1) = 0$ $x - 8 = 0$ or $x + 1 = 0$ $x = 8$ or $x = -1$	2. $x^2 - 4x + 4 = 0$
3. $x^2 - 8x = 0$	4. $4x^2 - 9 = 0$
5. $6x^2 + 13x - 5 = 0$	6. $(3x + 1)(x - 1) = 7$
7. $\frac{40+3x}{5+2x} = x$	8. $(x + 1)(x - 5) = 16$ $x^2 - 4x - 5 = 16$ $x^2 - 4x - 21 = 0$ $(x-7)(x-3) = 0$ $x = 7$ or $x = 3$

Exercise 3.1.1:

Solve the following quadratic equation by factorisation.

1. $x^2 - 5x - 6 = 0$ [6,-1]
2. $m^2 + 5m - 24 = 0$ [-8,3]
3. $y^2 + 10y + 24 = 0$ [-6,-4]
4. $2x^2 + 3x - 5 = 0$ $[1, -\frac{5}{2}]$
5. $16x^2 - 6x - 7 = 0$ $[\frac{7}{8}, -\frac{1}{2}]$
6. $2a^2 + 4a = 0$ [0.-2]
7. $100 - 9n^2 = 0$ $[\frac{10}{3}, -\frac{10}{3}]$
8. $(2x + 1)(x + 3) = 0$ $[-\frac{1}{2}, 3]$
9. $x^2 - 9x + 20 = 0$ [5,4]
10. $4x^2 - 13x + 3 = 0$ $[\frac{1}{4}, 3]$
11. $2x^2 - 3 = 5x$ $[-\frac{1}{2}, 3]$
12. $6x^2 - 11x = 7$ $[-\frac{1}{2}, \frac{7}{3}]$
13. $(2x - 3)^2 = 49$ [5,-2]
14. $(3m + 1)(m - 1) = 7$ $[2, -\frac{4}{3}]$
15. $10x^2 + 4 = 13x$ $[\frac{1}{2}, \frac{4}{5}]$
16. $x(x + 4) = 21$ [-7,3]

3.1.2 Solution by completing the square

By adding and subtracting a suitable constant, we put the x^2 and x terms in the quadratic equation so that they become complete square, and solve for x.

We can convert any quadratic equation to the form $(x + a)^2 — b^2 = 0$ and then we can easily find its roots.

1. The expression $x^2 - 2x + 1$ can be written in the form $(x - 1)^2$
   This is called "perfect square".

Example

Solve each of the following quadratic equation

(a) $(x + 1)^2 = 9$

$x + 1 = \pm 3$

$x + 1 = 3$, $x + 1 = -3$

$x = 2$, $x = -4$

(b) $x^2 = 49$

(c) $(x + 2)^2 = 36$

2. From the example , note that, if the algebraic expression on the LHS of the quadratic equation are perfect squares , the roots can be easily obtained by finding the square roots.

3. To make any quadratic expression $x^2 + hx$ into a perfect square , we add the term $(\frac{h}{2})^2$ to the expression .

   And this will make $x^2 + hx = x^2 + hx + \left(\frac{h}{2}\right)^2 = \left(x + \frac{h}{2}\right)^2$

4. To solve the equation by using completing the square method for quadratic equation $ax^2 + hx + k = 0$, follow this steps ;

   Step 1 : Rewrite the equation in the form $ax^2 + hx = -k$

   Step 2 : If the coefficient of x^2 is $\neq 1$, reduce the coefficient to 1 (by dividing) .

   Step 3 : Add $(\frac{h}{2})^2$ to both sides of the equation.

   Step 4 : Write the expression on the LHS as perfect square.

   Step 5 : Solve the equation

Examples

1. $x^2 + 6x - 9 = 0$

$x^2 + 6x = 9$

$x^2 + 6x + \left(\frac{6}{2}\right)^2 = 9 + \left(\frac{6}{2}\right)^2$

$(x+3)^2 = 18$

$x+3 = \sqrt{18}$

$x+3 = \pm 4.243$

$x = 4.243 - 3 \quad , \quad x = -4.243 - 3$

$x = 1.243 \quad , \quad x = -7.243$

2. $2x^2 - 5x - 8 = 0$

Exercise 3.1.2:

Solve the following equations by completing the square. (Give your answers correct to four significant figures)

1.	$x^2 - 8x + 14 = 0$	[5.41 , 2.59]
2.	$2x^2 - 7x - 1 = 0$	[3.64 , -0.14]
3.	$x^2 + 5x + 1 = 0$	[-0.209,-4.79]
4.	$-x^2 - 3x + 5 = 0$	[-4.19,1.19]
5.	$x^2 = 5(x+4)$	[7.62 , -2.62]
6.	$-4x^2 - 12x + 3 = 0$	[-3.23,0.232]
7.	$2x^2 - 3x - 4 = 0$	[2.35,-0.85]

3.1.3 Solution by quadratic Formula

- The root of equation also can be found by using the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- The discriminant $b^2 - 4ac > 0$ is positive when there are two distinct real solutions (or roots).
- The discriminant $b^2 - 4ac = 0$ when the two roots are equal.
- The discriminant $b^2 - 4ac < 0$ is negative when there are no real solutions (roots)

Example

$2x^2 - 7x - 3 = 0$

$a = 2 , b = -7 , c = -3$

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(2)(-3)}}{2(2)}$$

$$x = \frac{7 \pm \sqrt{73}}{4} = \frac{7 \pm 8.5440}{4}$$

$x = 3.886 , -0.386$

Exercise 3.1.3:

Use the quadratic formula to find the solutions of the following equations. Give your answers correct to three decimal places .

1. $x^2 - 3x - 5 = 0$ [4.193 , -1.193]
2. $9x^2 = 24x - 16$ [1.333]
3. $2x^2 + 5x - 1 = 0$ [0.186 , -2.686]
4. $3x^2 + 14x - 9 = 0$ [2.899 , -6.899]
5. $7 + 5x - x^2 = 0$ [0.768 , -0.434]
6. $m^2 = 20 - 4m$ [0.573 , -5.239]
7. $\frac{k+1}{3} = k^2$ [-1.140 , 6.140]
8. $x(x+4) = 3$ [0.646 , -4.646]

3.2 Simultaneous equation

There are two methods of solving simultaneous equations:

- By elimination
- By substitution

  Or, you can use calculator straight away to find the values of the unknowns.

3.2.1 Linear simultaneous equation in two unknowns

Exercise 3.2:

a) $\begin{aligned} x + 2y &= -1 \\ 4x - 3y &= 18 \end{aligned}$

b) $\begin{aligned} 3x - 2y &= 12 \\ x + 3y &= -7 \end{aligned}$

c) $\begin{aligned} 5m - 3n &= 11 \\ 3m + n &= 8 \end{aligned}$

d) $\begin{aligned} 3s + 2t &= 12 \\ 4s - t &= 5 \end{aligned}$

e) $\begin{aligned} 2x + 5y &= 7 \\ x + 3y &= 4 \end{aligned}$

f) $\begin{aligned} 3x - 2y &= 13 \\ 2x + 5y &= -4 \end{aligned}$

g) $\begin{aligned} 3x + 4y &= 5 \\ 2x - 5y &= -12 \end{aligned}$

h) $\begin{aligned} 7x - 2y &= 26 \\ 6x + 5y &= 29 \end{aligned}$

3.2.2 Linear simultaneous equation containing three unknowns

Systems of equations with three variables are only slightly more complicated to solve than those with two variables. The two most straightforward methods of solving these types of equations are by elimination and by using 3×3 matrices.

To use elimination to solve a system of three equations with three variables, follow this procedure:

1. **Write all the equations in standard form cleared of decimals or fractions.**
2. **Choose a variable to eliminate; then choose any two of the three equations and eliminate the chosen variable.**
3. **Select a different set of two equations and eliminate the same variable as in Step 2.**
4. **Solve the two equations from steps 2 and 3 for the two variables they contain.**
5. **Substitute the answers from Step 4 into any equation involving the remaining variable.**
6. **Check the solution with all three original equations.**

Example 1:

Solve this system of equations using elimination.

$$\begin{cases} 4x-2y+3z=1 & (1) \\ x+3y-4z=-7 & (2) \\ 3x+y+2z=5 & (3) \end{cases}$$

All the equations are already in the required form.

Choose a variable to eliminate, say x, and select two equations with which to eliminate it, say equations (1) and (2).

$$\begin{aligned} 4x-2y+3z=1 &\rightarrow \quad 4x-2y+3z=1 \\ x+3y-4z=-7 &\xrightarrow{\text{multiply } (-4)} \quad \underline{-4x-12y+16z=28} \\ &\qquad\qquad -14y+19z=29 \quad (4) \end{aligned}$$

Select a different set of two equations, say equations (2) and (3), and eliminate the same variable.

$$\begin{aligned} x+3y-4z=-7 &\xrightarrow{\text{multiply } (-3)} -3x-9y+12z=21 \\ 3x+y+2z=5 &\xrightarrow{\text{multiply } (-3)} \quad \underline{3x+y+2z=5} \\ &\qquad\qquad -8y+14z=26 \quad (5) \end{aligned}$$

Solve the system created by equations (4) and (5).

$$\begin{aligned} -14y+19z &= 29 \quad \overset{\text{multiply } (-8)}{\rightarrow} \quad 112y-152z = -232 \\ -8y+14z &= 26 \quad \overset{\text{multiply } (14)}{\rightarrow} \quad \underline{-112y+196z = \ 364} \\ 44z &= \ 132 \\ z &= \ 3 \end{aligned}$$

Now, substitute $z = 3$ into equation (4) to find y.

$$\begin{aligned} -14y-19z &= 29 \\ -14y-19(3) &= 29 \\ -14y+57 &= 29 \\ -14y &= -28 \\ y &= 2 \end{aligned}$$

Use the answers from Step 4 and substitute into any equation involving the remaining variable.

Using equation (2),

$$\begin{aligned} x+3y-4z &= -7 \\ x+3(2)-4(3) &= -7 \\ x+6-12 &= -7 \\ x &= -1 \end{aligned}$$

The solution is $x = -1$, $y = 2$, $z = 3$.

Example 2:

Solve this system of equations using the elimination method.

$$\begin{cases} x = 3z-5 & (1) \\ 2x+2z = y+16 & (2) \\ 7x-5z = 3y+19 & (3) \end{cases}$$

Write all equations in standard form.

$$\begin{aligned} x - 3z &= -5 \quad (1) \\ 2x - y + 2z &= 16 \quad (2) \\ 7x-3y-5z &= 19 \quad (3) \end{aligned}$$

Notice that equation (1) already has the y eliminated. Therefore, use equations (2) and (3) to eliminate y. Then use this result, together with equation (1), to solve for x and z. Use these results and substitute into either equation (2) or (3) to find y.

$$\begin{aligned} 2x - y + 2z &= 16 \quad \overset{\text{multiply } (-3)}{\rightarrow} \quad -6x+3y-6z = -48 \\ 7x-3y-5z &= 19 \quad \rightarrow \quad \underline{7x-3y-5z = \ 19} \\ x - 11z &= -29 \end{aligned}$$

$$\begin{aligned} x - 3z &= -5 \quad \overset{\text{multiply } (-1)}{\rightarrow} \quad -x + 3z = 5 \\ x - 11z &= -29 \quad \rightarrow \quad \underline{x - 11z = -29} \\ -8z &= -24 \\ z &= 3 \end{aligned}$$

Substitute $z = 3$ into equation (1).

$$\begin{aligned} x - 3z &= -5 \\ x - 3(3) &= -5 \\ x - 9 &= -5 \\ x &= 4 \end{aligned}$$

Substitute $x = 4$ and $z = 3$ into equation (2).

$$\begin{aligned} 2x - y - 2z &= 16 \\ 2(4) - y + 2(3) &= 16 \\ 8 - y + 6 &= 16 \\ -y &= 2 \\ y &= -2 \end{aligned}$$

The solution is $x = 4$, $y = -2$, $z = 3$.

Exercise 3.2.2:

Solve the simultaneous equations

1. $x - y + 3z = 5$
   $x + y + 6z = 12$
   $3x - 2y + 2z = 10$

2. $2x + 3y + 5z = 21$
   $3x + 6y + 15z = 51$
   $5x + 4y + 10z = 37$

3. $2x + 4y - z = 15$
   $3x + 8y + z = 44$
   $x + 2y + 2z = 15$

3.2.3 Simultaneous equation in two unknown involving one linear equation and one quadratic equation.

- Make one of the unknowns the subject of the linear equation (rearranging where necessary).
- Use the linear equation to substitute into the quadratic equation.
- There are usually two pairs of solutions.

Example 1 Solve the simultaneous equations $y = x + 1$ and $x^2 + y^2 = 13$

$x^2 + (x + 1)^2 = 13$	**1** Substitute $x + 1$ for y into the second equation.
$x^2 + x^2 + x + x + 1 = 13$ $2x^2 + 2x + 1 = 13$	**2** Expand the brackets and simplify.
$2x^2 + 2x - 12 = 0$ $(2x - 4)(x + 3) = 0$ So $x = 2$ or $x = -3$	**3** Factorise the quadratic equation. **4** Work out the values of x.
Using $y = x + 1$ When $x = 2$, $y = 2 + 1 = 3$ When $x = -3$, $y = -3 + 1 = -2$	**5** To find the value of y, substitute both values of x into one of the original equations.
So the solutions are $x = 2$, $y = 3$ and $x = -3$, $y = -2$	
Check: equation 1: $3 = 2 + 1$ YES and $-2 = -3 + 1$ YES equation 2: $2^2 + 3^2 = 13$ YES and $(-3)^2 + (-2)^2 = 13$ YES	**6** Substitute both pairs of values of x and y into both equations to check your answers.

Example 2 Solve $2x + 3y = 5$ and $2y^2 + xy = 12$ simultaneously.

$x = \frac{5-3y}{2}$	**1** Rearrange the first equation.
$2y^2 + \left(\frac{5-3y}{2}\right)y = 12$	**2** Substitute $\frac{5-3y}{2}$ for x into the second equation. Notice how it is easier to substitute for x than for y.
$2y^2 + \frac{5y-3y^2}{2} = 12$ $4y^2 + 5y - 3y^2 = 24$	**3** Expand the brackets and simplify.
$y^2 + 5y - 24 = 0$ $(y + 8)(y - 3) = 0$ So $y = -8$ or $y = 3$	**4** Factorise the quadratic equation. **5** Work out the values of y.
Using $2x + 3y = 5$ When $y = -8$, $2x + 3 \times (-8) = 5$, $x = 14.5$ When $y = 3$, $2x + 3 \times 3 = 5$, $x = -2$	**6** To find the value of x, substitute both values of y into one of the original equations.
So the solutions are $x = 14.5$, $y = -8$ and $x = -2$, $y = 3$	
Check: equation 1: $2 \times 14.5 + 3 \times (-8) = 5$ YES and $2 \times (-2) + 3 \times 3 = 5$ YES equation 2: $2\times(-8)^2 + 14.5\times(-8) = 12$ YES and $2 \times (3)^2 + (-2) \times 3 = 12$ YES	**7** Substitute both pairs of values of x and y into both equations to check your answers.

Exercise 3.2.3:

Solve these simultaneous equations.

1 $y = 2x + 1$
$x^2 + y^2 = 10$

2 $y = 6 - x$
$x^2 + y^2 = 20$

3 $y = x - 3$
$x^2 + y^2 = 5$

4 $y = 9 - 2x$
$x^2 + y^2 = 17$

5 $y = 3x - 5$
$y = x^2 - 2x + 1$

6 $y = x - 5$
$y = x^2 - 5x - 12$

7 $y = x + 5$
$x^2 + y^2 = 25$

8 $y = 2x - 1$
$x^2 + xy = 24$

9 $y = 2x$
$y^2 - xy = 8$

10 $2x + y = 11$
$xy = 15$

11 $x - y = 1$
$x^2 + y^2 = 3$

12 $y - x = 2$
$x^2 + xy = 3$

Answer:

1 $x = 1, y = 3$

$x = -\frac{9}{5}, y = -\frac{13}{5}$

2 $x = 2, y = 4$

$x = 4, y = 2$

3 $x = 1, y = -2$

$x = 2, y = -1$

4 $x = 4, y = 1$

$x = \frac{16}{5}, y = \frac{13}{5}$

5 $x = 3, y = 4$

$x = 2, y = 1$

6 $x = 7, y = 2$

$x = -1, y = -6$

7 $x = 0, y = 5$

$x = -5, y = 0$

8 $x = -\frac{8}{3}, y = -\frac{19}{3}$

9 $x = -2, y = -4$

$x = 2, y = 4$

10 $x = \frac{5}{2}, y = 6$

$x = 3, y = 5$

11 $x = \frac{1+\sqrt{5}}{2}, y = \frac{-1+\sqrt{5}}{2}$

$x = \frac{1-\sqrt{5}}{2}, y = \frac{-1-\sqrt{5}}{2}$

12 $x = \frac{-1+\sqrt{7}}{2}, y = \frac{3+\sqrt{7}}{2}$

$x = \frac{-1-\sqrt{7}}{2}, y = \frac{3-\sqrt{7}}{2}$